

GRAPHS: ELECTRIC CURRENT AND RANDOM WALK INTERPRETATIONS

JOHN E. HUTCHINSON

ABSTRACT. We build up the basics of finite graphs and their interpretations via electric currents, random walks and matrices, sufficient to develop analysis on fractals.

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- The *weighted degree* of x is $c_x := \sum_y c_{xy}$. It is the number of edges from x if $\mathbb{G}$ is simple and in this case is just called the *degree* of x .
- The *total mass* of $\mathbb{G}$ is $C = C(\mathbb{G}) := \sum_x c_x$. In the case of a simple graph it is twice the number of edges, or equivalently the sum of the degrees of the vertices.

3. Electric Network Interpretation of a Graph

3.1. Physical Motivation. Fix the graph $\mathbb{G}$. Think of $\mathbb{G}$ as an electric network [EN] with resistance r_{xy} on each edge xy . Suppose $\emptyset \neq G \subset V$ and $a \in V \setminus G$. Voltages $v(a) \neq 0$ and $v(b) = 0$ for $b \in G$ are imposed. (That is, the circuit is grounded on G .)

An important case is $G = \{b\}$ contains a single node. The set of nodes $\{a\} \cup G$ is the set of *boundary* nodes. Later we impose more general boundary conditions on v . The *interior* nodes are $V^\circ = V \setminus G$.

- Each node x settles into a steady state voltage $v(x)$.
- A *current* i_{xy} flows from x to y where

$$i_{xy} = (v_x - v_y)c_{xy}. \quad (\text{Ohm's law})$$

Thus $i_{xy} = -i_{yx}$ and current flows “downhill”.

- For x an interior node there is *zero net current* flowing out of x , i.e. into $\mathbb{G}$ through x :

$$\sum_y i_{xy} \left(\text{i.e. } \sum_y (v_x - v_y)c_{xy} \right) = 0. \quad (\text{Kirchoff's law})$$

3.2. Mathematical Framework. Because of the previous physical motivation¹ define the *voltage* to be the unique (see below) function v with²

$$(1) \quad \text{prescribed } v(a) > 0, \quad v(b) = 0 \text{ for } b \in G,$$

$$(2) \quad v(x) = \frac{1}{c_x} \sum_{y \sim x} c_{xy} v(y), \quad x \neq a, x \notin G.$$

Thus for interior x , $v(x)$ is a *weighted average* of v at neighbouring nodes.

Note that the second set of boundary conditions and (2) are *homogeneous*, i.e. preserved under addition and scalar multiplication of functions.

The following are true for general boundary voltages, with the same proofs.

- Denote the *Laplacian* of $f : V \rightarrow \mathbb{R}$ by³

$$(3) \quad \Delta f(x) := \left(\frac{1}{c_x} \sum_{y \sim x} c_{xy} f(y) \right) - f(x) = \frac{1}{c_x} \sum_{y \sim x} c_{xy} (f(y) - f(x)).$$

Thus (2) implies

$$(4) \quad \Delta v(x) = 0, \quad x \neq a, x \notin G.$$

For this reason we say v is *harmonic* at interior nodes.

- *Maximum Principle:* For interior x ; $v(x) \leq \max_{y \sim x} v(y)$ by (2). Thus

$$(5) \quad 0 \leq v(x) \leq v(a) \text{ for all } x.$$

- *Uniqueness:* Any two solutions of (1) and (2) are equal. (Apply (5) to the difference of the solutions.)
- *Existence:* There *exists* a (unique) solution of (2). (Since uniqueness implies existence for n linear equations in n variables.)
- The *current* along the (directed) edge xy is $i_{xy} := (v_x - v_y)c_{xy}$. Clearly $i_{xy} = -i_{yx}$.

¹Equation (2) is a rewriting of Kirchoff's law at x .

²We could as well take $v(a) < 0$, but it is convenient to take a positive voltage at a as we then have a current flowing from a into $\mathbb{G}$ and then out of $\mathbb{G}$ through G .

³So $\Delta f(a)$ is the difference between the weighted average of f at neighbours of a and the value $f(a)$ itself.

- The *current out of* $x \in V$, or *into* $\mathbb{G}$ at x , is defined by

$$(6) \quad i_x := \sum_y i_{xy} = -c_x \Delta v(x).$$

Thinking of the case $i_x < 0$, $-i_x$ is the current *out of* $\mathbb{G}$ at x .

Note $i_x = 0$ at interior nodes by (4). The *flux* of i through the network is⁴

$$(7) \quad F(i) := i_a = -i_b.$$

3.3. Rescaling Solutions. We make a simple but very useful observation which depends essentially on the homogeneity in (1) and (2).

If f_1 and f_2 are the solutions of (1) and (2) with prescribed $f_1(a)$ and $f_2(a)$, then trivially by uniqueness and homogeneity, they are equal up to rescaling:

$$(8) \quad f_1(x) = \frac{f_1(a)}{f_2(a)} f_2(x), \quad \forall x.$$

Moreover,

$$(9) \quad \frac{f_1(a)}{f_2(a)} = \frac{\Delta f_1(a)}{\Delta f_2(a)}.$$

So either of the above can be used as the scaling factor in (8).

3.4. Effective Conductance and Resistance. The important case is $G = \{b\}$.

- Motivated by Ohm's law,⁵ the *effective conductance* and *effective resistance* between a and G are defined by

$$(10) \quad \mathbb{C}_{aG} = \frac{1}{\mathbb{R}_{aG}} = \frac{i_a}{v(a)}.$$

Note $\mathbb{C}_{aG}$ and $\mathbb{R}_{aG}$ are independent of v by (8).

- *Symmetry*:⁶

$$\mathbb{C}_{ab} = \mathbb{C}_{ba}, \quad \mathbb{R}_{ab} = \mathbb{R}_{ba}.$$

- *Triangle Inequality*: The effective resistance defines a metric on $\mathbb{G}$ since

$$(11) \quad \mathbb{R}_{ac} \leq \mathbb{R}_{ab} + \mathbb{R}_{bc}.$$

Proof. ⁷ Let v_1 be the voltage function defined as the harmonic extension of $v_1(a) = \mathbb{R}_{ab}$ and $v_1(b) = 0$. Let i_1 denote the corresponding current, a unit flow from a to b .

Let v_2 be the voltage function defined as the harmonic extension of $v_2(b) = 0$ and $v_2(c) = -\mathbb{R}_{ac}$. Let i_2 denote the corresponding current, which in particular is a unit flow from b to c .

Let $v = v_1 + v_2$ and so the corresponding current is $i = i_1 + i_2$. Then i is a unit flow from a to c in the sense of (45), since $i(a) = i_1(a) + i_2(a) = 1 + 0 = 1$, $i(b) = i_1(b) + i_2(b) = -1 + 1 = 0$ and $i(c) = i_1(c) + i_2(c) = 0 - 1 = -1$. Notice that v is harmonic at b since

$$\Delta v(b) = \Delta v_1(b) + \Delta v_2(b) = -c_b^{-1}(i_1(b) + i_2(b)) = 0.$$

It follows that v is the harmonic extension of $v(a) = \mathbb{R}_{ab} + v_2(a)$ and of $v(c) = v_1(c) - \mathbb{R}_{ac}$. Since i is a *unit* flow from a to c it follows that $v(a) - v(c) = \mathbb{R}_{ac}$.

⁴Equality comes from

$$0 = \sum_x \sum_y i_{xy} = \sum_y i_{ay} + \sum_y i_{by} = i_a + i_b,$$

where the first equality is because $i_{xy} = -i_{yx}$ and the second because $i_x = 0$ at interior nodes.

⁵Suppose a wire of conductance $\mathbb{C}_{ab}$ connects a and b and a voltage difference $v(a)$ is applied across the wire. Then the current i_a that flows satisfies $\mathbb{C}_{ab} = i_a/v(a)$.

⁶If f and g are harmonic so is the linear combination $h = \alpha f + \beta g$ (*superposition principle*). The current corresponding to h along an edge is then the same linear combination of the currents corresponding to f and g along the edge. If f is defined by $f(a) = 1$ and $f(b) = 0$ then the current out of a is $\mathbb{C}_{ab}$. If g is defined by $g(a) = 0$ and $g(b) = 1$ then the current out of b is $\mathbb{C}_{ba}$, and hence out of a is $-\mathbb{C}_{ba}$ by (7). It follows that the current corresponding to $h = f + g$ and out of a is $\mathbb{C}_{ab} - \mathbb{C}_{ba}$. This equals 0 since $h(x) = 1 \forall x$.

⁷I haven't seen this proof before. It is elementary and gives a precise value for the error term in the triangle inequality.

But $v_2(a) \leq 0$ and $v_1(c) \geq 0$ by the maximum principle, and so

$$\mathbb{R}_{ac} = v(a) - v(c) = \mathbb{R}_{ab} + \mathbb{R}_{bc} + v_2(a) - v_1(c) \leq \mathbb{R}_{ab} + \mathbb{R}_{bc}.$$

□

We give a random walk proof on page 13 and a proof using trace networks on *** page ?? ***.

3.5. Computing Effective Conductance.

3.5.1. *Resistances in parallel and in series.* Conductances add in parallel, resistances add in series.

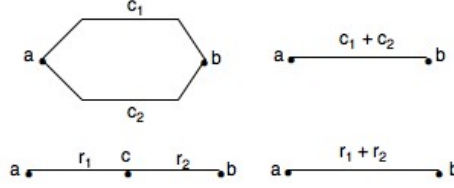


FIGURE 2. Conductances add in parallel, resistances add in series.

In Figure 2 voltages $v(a), v(b)$ are prescribed, as are conductances c_1, c_2 and resistances r_1, r_2 . In the top two diagrams the current flowing from a to b is clearly

$$(v(a) - v(b))(c_1 + c_2).$$

In the bottom left diagram direct calculation shows the induced voltage at c is

$$\frac{\frac{1}{r_1}v(a) + \frac{1}{r_2}v(b)}{\frac{1}{r_1} + \frac{1}{r_2}}.$$

Easy calculation then shows the current flowing out of a , and the current flowing into b , is

$$\frac{v(a) - v(b)}{r_1 + r_2}$$

in both of the bottom diagrams.

It follows that in any circuit the two left side diagrams can be replaced by the corresponding right side diagrams when computing induced voltages, current flow or effective resistance. They have the same input-output relations on the nodes $\{a, b\}$ and so we say they are *equivalent* on the nodes $\{a, b\}$. See also Section 11.*** See [DS, p 44] for an example calculation.

3.5.2. *The Δ -Y transform.* Consider the two networks in Figure 3 with conductances as shown.

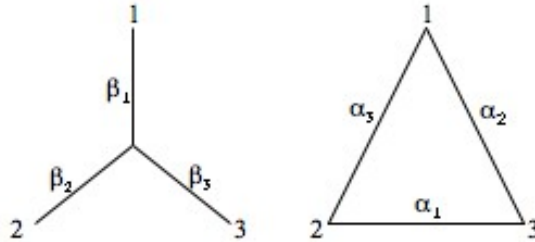


FIGURE 3. The Δ -Y transform

We say they are *equivalent* if they have the same input output relations on the nodes $\{1, 2, 3\}$.

Proposition 3.1. *The networks in Figure 3 are equivalent on $\{1, 2, 3\}$ iff*

$$(12) \quad \alpha_1 = \frac{\beta_2\beta_3}{\beta_1 + \beta_2 + \beta_3}, \quad \alpha_2 = \frac{\beta_3\beta_1}{\beta_1 + \beta_2 + \beta_3}, \quad \alpha_3 = \frac{\beta_1\beta_2}{\beta_1 + \beta_2 + \beta_3}.$$

Equivalently, iff

$$(13) \quad \beta_i = \frac{S}{\alpha_i}, \text{ where } S = \alpha_1\alpha_2 + \alpha_2\alpha_3 + \alpha_3\alpha_1.$$

Proof. ⁸ It is sufficient to consider a voltage 1 at node 1 and 0 at nodes 2 and 3, since by symmetry this covers two similar cases by rotating, and then all cases by scaling and summing boundary data.

In the Y diagram, the induced voltage at the central node is $\beta_1/(\beta_1 + \beta_2 + \beta_3)$. The currents through 1, 2 and 3 then are:

$$i_1 = \beta_1 \left(1 - \frac{\beta_1}{\beta_1 + \beta_2 + \beta_3}\right) = \frac{\beta_1(\beta_2 + \beta_3)}{\beta_1 + \beta_2 + \beta_3}, \quad i_2 = \frac{\beta_1\beta_2}{\beta_1 + \beta_2 + \beta_3}, \quad i_3 = \frac{\beta_1\beta_3}{\beta_1 + \beta_2 + \beta_3}$$

In the Δ diagram the corresponding currents are

$$i_1 = \alpha_3 + \alpha_2, \quad i_2 = \alpha_3, \quad i_3 = \alpha_2.$$

Hence

$$\alpha_2 = \frac{\beta_1\beta_3}{\beta_1 + \beta_2 + \beta_3}, \quad \alpha_3 = \frac{\beta_1\beta_2}{\beta_1 + \beta_2 + \beta_3},$$

and so

$$\alpha_1 = \frac{\beta_2\beta_3}{\beta_1 + \beta_2 + \beta_3}$$

by symmetry.

It follows that

$$S := \alpha_1\alpha_2 + \alpha_2\alpha_3 + \alpha_3\alpha_1 = \frac{\beta_1\beta_2\beta_3(\beta_1 + \beta_2 + \beta_3)}{(\beta_1 + \beta_2 + \beta_3)^2} = \frac{\beta_1\beta_2\beta_3}{\beta_1 + \beta_2 + \beta_3},$$

and so $\beta_i = S/\alpha_i$. □

Alternative Proof. From the conductance matrix for Y ,

$$A = \begin{bmatrix} -\beta_1 & 0 & 0 & \beta_1 \\ 0 & -\beta_2 & 0 & \beta_2 \\ 0 & 0 & -\beta_3 & \beta_3 \\ \beta_1 & \beta_2 & \beta_3 & -\beta_1 - \beta_2 - \beta_3 \end{bmatrix}$$

one can use Maple to compute the trace conductance matrix B on Δ via (55). Namely,

$$B = A_{GG} - A_{GH}A_{HH}^{-1}A_{HG} = \begin{bmatrix} -\frac{\beta_1(\beta_2+\beta_3)}{\beta_1+\beta_2+\beta_3} & \frac{\beta_1\beta_2}{\beta_1+\beta_2+\beta_3} & \frac{\beta_1\beta_3}{\beta_1+\beta_2+\beta_3} \\ \frac{\beta_1\beta_2}{\beta_1+\beta_2+\beta_3} & -\frac{\beta_2(\beta_1+\beta_3)}{\beta_1+\beta_2+\beta_3} & \frac{\beta_2\beta_3}{\beta_1+\beta_2+\beta_3} \\ \frac{\beta_1\beta_3}{\beta_1+\beta_2+\beta_3} & \frac{\beta_2\beta_3}{\beta_1+\beta_2+\beta_3} & -\frac{\beta_3(\beta_1+\beta_2)}{\beta_1+\beta_2+\beta_3} \end{bmatrix}.$$

□

4. Random Walk Interpretation of a Graph

4.1. Definition of the RW on $\mathbb{G}$. This is characterised by defining the probability of moving from x to y to be

$$(14) \quad p_{xy} = \frac{c_{xy}}{c_x}.$$

Thus rows of the transition matrix $P := (p_{xy})_{x \in V, y \in V}$ are probability vectors, with row x containing the probabilities corresponding to leaving node x .

The RW always begins at time 0, and its position at time $n \geq 0$ is denoted by X_n .

⁸This is less tedious than computing the effective resistances between the pairs 12, 23 and 31.

4.2. Markov Chain. The random walk is a markov chain with state space V and transition probabilities p_{xy} .

- The markov chain is *irreducible* since $\mathbb{G}$ is connected, and so any state can be reached from any other state.
- Hence by standard markov chain theory there is a *unique steady state distribution* $(w_x)_{x \in V}$.
- As for any markov chain, if $(u_x)_{x \in V}$ is the probability distribution at one time then $(v_x)_{x \in V}$, where $v_x = \sum_y u_y p_{yx}$, gives the probability distribution at the next time.
- It follows by direct calculation that the *steady state distribution* w_x is *proportional to* c_x ,⁹ and so

$$(15) \quad w_x = \frac{c_x}{C}.$$

- The markov chain is *reversible*, i.e.

$$w_x p_{xy} = w_y p_{yx} \quad \forall x, y \in V.$$

- *Interpretation of reversibility:* One cannot tell the direction of time by watching the RW evolve from the steady state distribution w . More precisely

$$\begin{aligned} P(xy) &:= P(X_0 = x) P(X_1 = y | X_0 = x) = w_x p_{xy} \\ &= w_y p_{yx} = P(X_0 = y) P(X_1 = x | X_0 = y) =: P(yx). \end{aligned}$$

Similarly

$$\begin{aligned} P(x_0 x_1 \dots x_n) &= w_{x_0} p_{x_0 x_1} p_{x_1 x_2} \dots p_{x_{n-1} x_n} \\ &= w_{x_n} p_{x_n x_{n-1}} \dots p_{x_2 x_1} p_{x_1 x_0} = P(x_n \dots x_1 x_0). \end{aligned}$$

The following equivalent version does not explicitly refer to the steady state distribution. For any path $x_0 x_1 \dots x_n$ and beginning at each end respectively,

$$c_{x_0} P^{x_0}(x_0 x_1 \dots x_n) = c_{x_n} P^{x_n}(x_n \dots x_1 x_0).$$

- Conversely, any reversible markov chain is the RW given by a graph, provided we allow $c_{xx} > 0$.¹⁰
- A natural measure on V is $(c_x)_{x \in V}$. The corresponding *transition density* is defined for $x \sim y$ by

$$p(x, y) := \frac{P^x(X_1 = y)}{c_y} = \frac{c_{xy}}{c_x c_y} = \frac{P^y(X_1 = x)}{c_x} = p(y, x).$$

Reversibility thus is equivalent to symmetry of the transition density $p(x, y)$ (but *not* of p_{xy}).

4.3. Two functions associated with the RW. Suppose $\emptyset \neq G \subset V$, $a \in V \setminus G$ and $x \in V$.

We introduce two important functions h and g which equal 0 on G and are harmonic at $x \notin \{a\} \cup G$. By the scaling comments in Section 3.3, they are multiples of each other, of the voltage v in (1) and (2), and the Green's function discussed later. We note the consequences in Section 5.

Let

$$(16) \quad h(x) = P^x(T_a < T_G),$$

$$(17) \quad g(x) = \frac{1}{c_x} \mathbb{E}^a(V_{x \rightarrow G}).$$

⁹Since if $w_x := c_x/C$ then

$$\sum_y w_y p_{yx} = \sum_y \frac{c_y}{C} \frac{c_{yx}}{c_y} = \sum_y \frac{c_{yx}}{C} = \sum_y \frac{c_{xy}}{C} = \frac{c_x}{C} = w_x.$$

Hence w_x is the steady state distribution by uniqueness.

¹⁰Suppose the markov chain is p and the steady state distribution is w . Then $w_x p_{xy} = w_y p_{yx} \forall x, y$. Define $c_{xy} = w_x p_{xy}$ and note $c_{xy} = c_{yx}$. Moreover,

$$\frac{c_{xy}}{c_x} = \frac{c_{xy}}{\sum_z c_{xz}} = \frac{w_x p_{xy}}{\sum_z w_x p_{xz}} = p_{xy},$$

and so this gives the original markov chain.

By T_a (T_G) is meant *the time of first arrival at a G* — where here the walk starts from x .

By $V_{x \rightarrow G}$ is meant *the number of visits to x before arriving at G* — where here the walk starts from a . If $x = a$ then *the initial visit at time 0 is included* as a visit. If $x \in G$ then $V_{x \rightarrow G} = 0$.

Proposition 4.1.

$$(18) \quad \begin{aligned} h(a) &= 1, & h(b) &= 0 \text{ if } b \in G, \\ g(a) &= \frac{1}{c_a} \mathbb{E}^a V_{a \rightarrow b}, & g(b) &= 0 \text{ if } b \in G. \end{aligned}$$

Moreover,

$$(19) \quad \Delta h(x) = 0, \quad \Delta g(x) = 0, \quad x \notin \{a\} \cup G.$$

Proof. The boundary conditions are immediate.

Because the start from x is *followed* by a visit to some $y \sim x$,

$$(20) \quad h(x) = \sum_y p_{xy} P^y(T_a < T_G) = \sum_y \frac{c_{xy}}{c_x} h(y);$$

and because every visit to x is *preceded* by a visit to some $y \sim x$,¹¹

$$(21) \quad \begin{aligned} g(x) &:= \frac{1}{c_x} \mathbb{E}^a V_{x \rightarrow G} = \frac{1}{c_x} \sum_y p_{yx} \mathbb{E}^a V_{y \rightarrow G} \\ &= \sum_y \frac{1}{c_x} \frac{c_{yx}}{c_y} c_y g(y) = \sum_y \frac{c_{xy}}{c_y} g(y). \end{aligned}$$

□

Note that *reversibility is required for the proof that g is harmonic*, but not so for h .

For later purposes, keeping in mind the scaling comments in Section 3.3, we next compute the Laplacians at a .

Let T_a^+ denotes the first *return* time to a , i.e. the time of the first visit to a *after* time 0 or equivalently the first visit in positive time. Then

$$(22) \quad P^a(T_G < T_a^+)$$

denotes the probability the walk reaches G from a without returning to a . It is also called the *escape probability* from a to G .

Proposition 4.2.

$$(23) \quad -\Delta h(a) = P^a(T_G < T_a^+), \quad -\Delta g(a) = \frac{1}{c_a}.$$

Proof. For h ,

$$\begin{aligned} -\Delta h(a) &= \sum_x (h(a) - h(x)) p_{ax} \\ &= \sum_x (1 - P^x(T_a < T_G)) p_{ax} \\ &= \sum_x p_{ax} P^x(T_G < T_a) = P^a(T_G < T_a^+). \end{aligned}$$

¹¹More precisely, fix $x \notin \{a\} \cup G$. Then

$$\begin{aligned} g(x) &= \frac{1}{c_x} \mathbb{E}^a (V_{x \rightarrow G}) = \frac{1}{c_x} \sum_{n \geq 1} P^a(X_n = x) \quad (\text{as } x \neq a) \\ &= \frac{1}{c_x} \sum_{n \geq 1} \sum_{y \sim x} P^a(X_{n-1} = y) p_{yx} = \frac{1}{c_x} \sum_{y \sim x} \frac{c_{yx}}{c_y} \mathbb{E}^a (V_{y \rightarrow G}) = \sum_{y \sim x} \frac{c_{xy}}{c_x} g(y). \end{aligned}$$

For g use the same calculation as in (21) after noting

$$g(a) := \frac{1}{c_a} \mathbb{E}^a V_{a \rightarrow G} = \frac{1}{c_a} + \frac{1}{c_a} \sum_y p_{ya} \mathbb{E}^a V_{y \rightarrow G},$$

where the second equality is because every visit to a except for the first is preceded by a visit to some $y \sim a$. $\square$

We give a random walk proof of the following. It is also a consequence of the electric network interpretation; see (??) and (??). *** check references and claim ***

Proposition 4.3. *For $a \notin G$,*

$$(24) \quad \boxed{\mathbb{E}^a V_{a \rightarrow G} = 1/P^a(T_G < T_a^+).}$$

Proof. Let

$$\begin{aligned} p &= P^a(T_G < T_a^+) \quad (\text{i.e. "success"}), \\ 1 - p &= P^a(T_a^+ < T_G) \quad (\text{i.e. "failure"}). \end{aligned}$$

The number of “failures” before a success is a geometric distribution with parameter p . That is, the probability of exactly n returns to a before reaching G , or equivalently exactly $n + 1$ visits, is $(1 - p)^n p$. Hence the expected number of visits to a before reaching G is $1 +$ the mean of the geometric distribution, and so by a standard result¹² is $1 + (1 - p)/p = 1/p$. $\square$

5. Connection between EN, RW and Analytic Interpretations

Suppose $\emptyset \neq G \subset V$, $a \in V \setminus G$ and $x \in V$.

5.1. Green’s Function.

5.1.1. *Definition.* Motivated by the classical situation,¹³ *Green’s function centred at a* (for the set G) is the unique function $\mathcal{G}_a^G = \mathcal{G}_a : V \rightarrow \mathbb{R}$ given by

$$(25) \quad \boxed{-\Delta \mathcal{G}_a(x) = \frac{1}{c_a} \mathcal{X}_a(x), \quad x \in V \setminus G,}$$

$$\mathcal{G}_a(x) = 0 \quad x \in G,$$

where $\mathcal{X}_a$ is the characteristic function equal to 1 at a and 0 elsewhere. Uniqueness, and hence existence, follow as in Section 3.2. Note that $\frac{1}{c_a} \mathcal{X}_a$ is the natural choice as the “Dirac measure” centred at a since it is supported on a and integrates to 1 w.r.t. the natural measure $(c_x)_{x \in V}$.

¹²To see this directly let Z denote the number of visits to a before first reaching G . Then

$$\begin{aligned} \mathbb{E}(Z) &= \sum_{n \geq 0} (n+1)(1-p)^n p = -p \sum_{n \geq 0} \frac{d}{dp} (1-p)^{n+1} \\ &= -p \frac{d}{dp} \sum_{n \geq 0} (1-p)^{n+1} = -p \frac{d}{dp} \left(\frac{1-p}{p} \right) = \frac{1}{p}. \end{aligned}$$

¹³Suppose $a \in \Omega$. Then the Green function $\mathcal{G}_a^{\partial\Omega}(x) = \mathcal{G}_a(x)$ is characterised for $x \in \bar{\Omega}$ by

$$\begin{aligned} -\Delta \mathcal{G}_a(x) &= \delta_a(x) \quad \text{if } x \in \Omega, \quad (\text{Dirac delta measure, equality in distributional sense}) \\ \mathcal{G}_a(x) &= 0 \quad \text{if } x \in \partial\Omega. \end{aligned}$$

In particular,

$$\mathcal{G}_a(x) = \begin{cases} -\frac{\ln|x-a|}{2\pi} + \psi(x) & n = 2, \\ \frac{c_n}{|x-a|^{n-2}} + \psi(x) & n \geq 3, \quad (c_3 = \frac{1}{4\pi}) \end{cases}$$

where $\psi(x)$ is the unique harmonic function on Ω such that $\mathcal{G}_a(x) = 0$ if $x \in \partial\Omega$.

5.1.2. *Symmetry.* This is analogous to the proof in the classical setting.¹⁴

$$(26) \quad \boxed{\mathcal{G}_b(a) = \mathcal{G}_a(b).} \quad a, b \notin G$$

Proof. From Green's identity (41), which is just an algebraic calculation,

$$(\Delta \mathcal{G}_a, \mathcal{G}_b) = (\mathcal{G}_a, \Delta \mathcal{G}_b),$$

where each side is the natural inner product with the weighting c_x at x . The left side is $c_a \mathcal{G}_b(a) \frac{1}{c_a} = \mathcal{G}_b(a)$ and the right side is similarly $\mathcal{G}_a(b)$. $\square$

5.1.3. *Poisson's equation.* Analogously to the PDE setting,¹⁵ the solution of

$$(27) \quad \begin{aligned} -\Delta u(x) &= f(x) & x \in V \setminus G, \\ u(x) &= 0 & x \in G, \end{aligned}$$

is given by

$$(28) \quad \boxed{u(x) = \sum_{y \in V \setminus G} \mathcal{G}_y(x) f(y) c_y.}$$

Proof. If $x \in G$ then

$$\sum_{y \in V \setminus G} \mathcal{G}_y(x) f(y) c_y = 0,$$

and if $x \in V \setminus G$ then

$$-\Delta_x \left(\sum_{y \in V \setminus G} \mathcal{G}_y(x) f(y) c_y \right) = \sum_{y \in V^\circ} \frac{\mathcal{X}_y(x)}{c_y} f(y) c_y = f(x). \quad \square$$

5.2. **Summary of Connections so far.** We defined the following functions on V , harmonic for $x \notin G$:

$$\begin{aligned} v(x) &= \text{voltage given by (1), (2),} \\ h(x) &= P^x(T_a < T_G), \\ g(x) &= \frac{1}{c_x} \mathbb{E}^a V_{x \rightarrow G}, \\ \mathcal{G}_a^G(x) &= \text{Green function centred at } a \text{ given by (25).} \end{aligned}$$

¹⁴Symmetry of Green's function is shown for $a \neq b \in \Omega$ as follows:

$$\mathcal{G}_a(b) - \mathcal{G}_b(a) = \int_{\Omega} \mathcal{G}_b \Delta \mathcal{G}_a - \mathcal{G}_a \Delta \mathcal{G}_b = \int_{\partial \Omega} \mathcal{G}_b \frac{\partial \mathcal{G}_a}{\partial \nu} - \mathcal{G}_a \frac{\partial \mathcal{G}_b}{\partial \nu} = 0$$

The first and third equalities are from the definition of $\mathcal{G}_a$ and $\mathcal{G}_b$. The second is by Green's formula. Strictly, one should integrate over $\Omega \setminus (B_\epsilon(a) \cup B_\epsilon(b))$ and let $\epsilon \rightarrow 0$.

¹⁵The solution of

$$\begin{aligned} -\Delta u &= f & \text{in } \Omega \\ u &= 0 & \text{on } \partial \Omega \end{aligned}$$

is given by the following "weighted sum" of Green functions:

$$u(x) = \int_{\Omega} \mathcal{G}_y(x) f(y) dy.$$

In fact formally, and made precise by a careful limiting argument, one has

$$\int_{\Omega} \mathcal{G}_y(x) f(y) dy = \int_{\Omega} \Delta_x \mathcal{G}_y(x) f(y) dy = \int_{\Omega} \delta_x(y) \mathcal{G}_y(x) f(y) dy = f(x).$$

Moreover,

$$\begin{aligned} v(a) &= 1 & -c_a \Delta v(a) &= i_a = \mathbb{C}_{aG} \\ h(a) &= 1 & -c_a \Delta h(a) &= c_a P^a(T_a < T_a^+) \\ g(a) &= \frac{1}{c_a} \mathbb{E}^a V_{a \rightarrow G} & -c_a \Delta g(a) &= 1 \\ \mathcal{G}_a^G(a) &= (\text{no nice analytic expr}) & -c_a \Delta \mathcal{G}_a^G(a) &= 1 \end{aligned}$$

In particular,

$$(29) \quad \boxed{v(x) = P^x(T_a < T_G) = \mathbb{C}_{aG} \frac{\mathbb{E}^a V_{x \rightarrow G}}{c_x} = \mathbb{C}_{aG} \mathcal{G}_a^G(x).}$$

The other information we extract using the values of these functions and their Laplacians at a , together with the comments in Section 3.3, is

$$(30) \quad \boxed{\mathbb{R}_{aG} = \frac{1}{i_a} = \frac{1}{c_a P^a(T_a < T_a^+)} = \frac{\mathbb{E}^a V_{a \rightarrow G}}{c_a} = \mathcal{G}_a^G(a).}$$

Note that the third equality above follows directly from the fact that g and h are multiples of each other, and is independent of the direct proof in Proposition 4.3.

Consider a non empty set of “boundary” nodes $B = \partial V \subset V$. This generalises the situation in Sections 3, 4 and 5, where $\partial V = \{a, b\}$ or $\partial V = \{a\}$.

5.3. Currents and Traverses along Edges. There is also a connection between the current along an edge and the net number of traverses along the edge.

For this let $Tr_{xy \rightarrow G}$ denote the *net number of traverses from x to y before reaching G* .

Proposition 5.1. *Suppose a voltage is applied at a and zero voltage at G so that unit current flows into V at a . Then if i_{xy} denotes the (signed) current along the edge xy ,*

$$(31) \quad i_{xy} = \mathbb{E}^a Tr_{xy \rightarrow G}.$$

Proof. The voltage for unit current flow is $g(x)$ since both are characterised by $-c_a \Delta v(a) = -c_a \Delta g(a) = 1$, see (6) and (23). Hence

$$\begin{aligned} i_{xy} &= (g(x) - g(y)) c_{xy} \\ &= \frac{c_{xy}}{c_x} \mathbb{E}^a (V_{x \rightarrow G}) - \frac{c_{xy}}{c_y} \mathbb{E}^a (V_{y \rightarrow G}) = p_{xy} \mathbb{E}^a (V_{x \rightarrow G}) - p_{yx} \mathbb{E}^a (V_{y \rightarrow G}) \\ &= \mathbb{E}^a (\text{no. of traverses from } x \text{ to } y \text{ before } G) - \mathbb{E}^a (\text{no. of traverses from } y \text{ to } x \text{ before } G) \\ &= \mathbb{E}^a (\text{net no. of traverses from } x \text{ to } y \text{ before } G) = \mathbb{E}^a Tr_{xy \rightarrow G}. \quad \square \end{aligned}$$

Note that for and $x \sim y$,

$$(32) \quad g(x) := \frac{1}{c_x} \mathbb{E}^a V_{x \rightarrow G} = \frac{1}{c_{xy}} \mathbb{E}^a (\text{no. of traverses from } x \text{ to } y \text{ before } G).$$

The right side is the normalised no. of traverses in the direction xy , *not* the net no. of traverses.

5.4. Dirichlet Problem. Suppose $G \subset V$. (There is here no role played by a .)

Analogous to the classical P.D.E. problem¹⁶ suppose $h : G \rightarrow \mathbb{R}$ is prescribed. Then there exists a unique extension of h to $V \setminus G$ satisfying

$$(33) \quad \Delta h(x) = 0 \quad x \notin G.$$

Uniqueness, and hence existence, follow as in Section 3.2.

¹⁶Let $\Omega \subset \mathbb{R}^n$ be non empty bounded and open with a “nice” boundary $\partial\Omega$, e.g. C^1 . Suppose $h : \partial V \rightarrow \mathbb{R}$ is prescribed and also “nice”, e.g. C^1 . Then there is a unique harmonic extension of h to Ω .

RW Interpretation: (This is analogous to the case of Brownian Motion in the P.D.E. setting.¹⁷) The solution h is the expected “payoff”:

$$(34) \quad \boxed{h(x) = \mathbb{E}^x h(X_{T_G}).}$$

where $T_{\partial V}$ is the time of first arrival at ∂V . The fact that h agrees with the given boundary values, and is harmonic on $V \setminus G$, is given by the same argument as for Proposition 4.1.

EN Interpretation: It also follows as in Sections 3.1 and 3.2 that we can regard h as giving the voltage at each node in V for the prescribed voltages on G .

6. Crossing Time, Einstein Relation

6.1. **Crossing Time Theorem.** [Chandra et al, 1989]

$$(35) \quad \boxed{\mathbb{E}^a T_b + \mathbb{E}^b T_a = \mathbb{R}_{ab} C(\mathbb{G}).}$$

Proof. ¹⁸ Starting from any point a ,

$$T_b = \sum_x V_{x \rightarrow b}.$$

It follows from the harmonicity of g in Section ?? that

$$\begin{aligned} \mathbb{E}^a T_b &= \sum_x \mathbb{E}^a(V_{x \rightarrow b}) = \sum_x c_x \frac{\mathbb{E}^a(V_{x \rightarrow b})}{c_x} \\ &= \sum_x c_x (\text{harmonic ext } v(x) \text{ to } V \text{ of } v(a) = \frac{\mathbb{E}^a(V_{a \rightarrow b})}{c_a}, v(b) = 0) \\ &= \frac{\mathbb{E}^a(V_{a \rightarrow b})}{c_a} \sum_x c_x (\text{harmonic ext } v(x) \text{ to } V \text{ of } v(a) = 1, v(b) = 0) \\ &= \mathbb{R}_{ab} \sum_x c_x (\text{harmonic ext } v(x) \text{ to } V \text{ of } v(a) = 1, v(b) = 0). \end{aligned}$$

Similarly, using the previously proved $\mathbb{R}_{ab} = \mathbb{R}_{ba}$,

$$\mathbb{E}^b T_a = \mathbb{R}_{ab} \sum_x c_x (\text{harmonic ext } v(x) \text{ to } V \text{ of } v(a) = 0, v(b) = 1).$$

Adding, and using the uniqueness of harmonic extension,

$$\begin{aligned} \mathbb{E}^a T_b + \mathbb{E}^b T_a &= \mathbb{R}_{ab} \sum_x c_x (\text{harmonic ext } v(x) \text{ to } V \text{ of } v(a) = 1, v(b) = 1) \\ &= \mathbb{R}_{ab} \sum_x c_x = \mathbb{R}_{ab} C(\mathbb{G}). \end{aligned} \quad \square$$

6.2. **Einstein Relation.** The left side of (35) is the expected *commute time* from a to b and back, $\mathbb{R}_{ab}$ is the distance between a and b in the resistance metric, and $C = C(\mathbb{G})$ is the total mass of $\mathbb{G}$ (see page 3). This gives the multiplicative version of the *Einstein relation*:

$$(36) \quad \boxed{\text{Time} = \text{Distance} \times \text{Mass}.}$$

We often have a “scale” or parameter x , such as Euclidean distance, where for appropriate exponents

$$(37) \quad \text{Time} \sim x^{d_w}, \text{ Distance} \sim x^{d_r}, \text{ Mass} \sim x^{d_f}.$$

More precisely: “Time” is the commute time between points which are separated by Euclidean distance $\approx x$, “Distance” is the resistance metric distance between points which are separated by Euclidean distance $\approx x$, and “Mass” is the mass of a ball whose Euclidean diameter is $\approx x$.

¹⁷If X is Brownian motion on Ω and $T_{\partial\Omega}$ is the time of first reaching the boundary, then

$$\phi(x) = \mathbb{E}^x \phi(X_{T_{\partial\Omega}}).$$

¹⁸I havn’t seen quite this proof before, but it is natural with the current development.

The exponents are called the *walk dimension* d_w , the *resistance dimension* d_r , and the *fractal dimension* d_f .

From (36) and (37) one gets the usual form of the *Einstein relation*:

$$(38) \quad \boxed{d_w = d_r + d_f.}$$

6.3. Triangle Inequality. Here is a random walk proof of the triangle inequality for the resistance metric. This also gives the error term, see Footnote 19.

Proof. Clearly¹⁹

$$\mathbb{E}^a T_c \leq \mathbb{E}^a T_b + \mathbb{E}^b T_c.$$

Switching a and c and adding,

$$\mathbb{E}^a T_c + \mathbb{E}^c T_a \leq \mathbb{E}^a T_b + \mathbb{E}^b T_a + \mathbb{E}^b T_c + \mathbb{E}^c T_b.$$

Dividing both sides by $C(\mathbb{G})$ and using the crossing time inequality (35) gives

$$\mathbb{R}_{ac} \leq \mathbb{R}_{ab} + \mathbb{R}_{bc}. \quad \square$$

7. Dirichlet Energy

7.1. Definitions.

- For $f, g : V \rightarrow \mathbb{R}$ define the *Dirichlet Energy* of f , and corresponding bilinear form, by

$$(39) \quad \begin{aligned} \mathcal{E}(f) &= \frac{1}{2} \sum_{x,y} c_{xy} (f(x) - f(y))^2, \\ \mathcal{E}(f, g) &= \frac{1}{2} \sum_{x,y} c_{xy} (f(x) - f(y))(g(x) - g(y)). \end{aligned}$$

The $\frac{1}{2}$ arises since the energy is “sitting” on edges, but each edge appears twice in the sum.

- *EN Interpretation:* If f is a voltage then $i_{xy} = c_{xy}(f(x) - f(y))$ is the current along the directed edge xy and the first summand in (39),

$$(40) \quad c_{xy} (f(x) - f(y))^2 = i_{xy} (f(x) - f(y)) = i_{xy}^2 r_{xy},$$

is the “energy dissipation” on the edge xy . The Dirichlet energy $\mathcal{E}(f)$ is the “energy dissipation” of the network, or equivalently the energy required to keep the network flowing.

7.2. Green’s Identity. (This is analogous to the P.D.E. case.)

$$(41) \quad \mathcal{E}(f, g) = -(\Delta f, g) = -(f, \Delta g),$$

where $(h, k) = \sum_x h(x)k(x)c_x$ denotes the L^2 inner product with the natural measure $(c_x)_{x \in V}$.

In particular, Δ is self-adjoint.

Note that (41) expresses the energy $\mathcal{E}(f)$ as a sum of contributions at *nodes* rather than edges, and the contribution is 0 at any node where $\Delta f = 0$ or $f = 0$.

Moreover, suppose f and g are applied voltages on ∂V and i^g denotes the corresponding currents flowing into $\mathbb{G}$ on ∂V . Then (41) can be written

$$(42) \quad \mathcal{E}(f, g) = \sum_{x \in \partial V} f(x) i^g(x).$$

¹⁹Using an obvious notation, $T_c^a = T_b^a + T_c^b$ if $T_b^a \leq T_c^a$ and $T_c^a = T_b^a - T_b^c < T_b^a + T_c^b$ if $T_c^a < T_b^a$. So $T_c^a \leq T_b^a + T_c^b$ in both cases.

More precisely,

$$\begin{aligned} \mathbb{E}^a T_c &= P^a(T_b \leq T_c) \mathbb{E}^a(T_c \mid T_b \leq T_c) + P^a(T_c < T_b) \mathbb{E}^a(T_c \mid T_c < T_b) \\ &= P^a(T_b \leq T_c) (\mathbb{E}^a T_b + \mathbb{E}^b T_c) + P^a(T_c < T_b) (\mathbb{E}^a T_b + \mathbb{E}^b T_c - (\mathbb{E}^b T_c + \mathbb{E}^c T_b)) \\ &= (\mathbb{E}^a T_b + \mathbb{E}^b T_c) - P^a(T_c < T_b) \mathbb{E}_c^b, \end{aligned}$$

where $C_c^b = C_b^c := T_c^b + T_b^c$ is the *commute time* from b to c and back to b , and $\mathbb{E}_c^b := \mathbb{E} C_c^b$.

Proof of (41).

$$\begin{aligned}
\mathcal{E}(f, g) &= \frac{1}{2} \sum_{x,y} c_{xy} f(x) (g(x) - g(y)) - \frac{1}{2} \sum_{x,y} c_{xy} f(y) (g(x) - g(y)) \\
&= \frac{1}{2} \sum_{x,y} c_{xy} f(x) (g(x) - g(y)) + \frac{1}{2} \sum_{x,y} c_{yx} f(x) (g(x) - g(y)) \\
&= \sum_{x,y} c_{xy} f(x) (g(x) - g(y)) = - \sum_x f(x) \Delta g(x) c_x.
\end{aligned}$$

□

7.3. Thompson's First Principle. (This is essentially just a particular case of the standard Dirichlet principle.)

Proposition 7.1. *Suppose $h|_{\partial V}$ is prescribed and $\Delta h = 0$ on V° . Then*

$$(43) \quad \mathcal{E}(h) = \min\{\mathcal{E}(f) : f = h \text{ on } \partial V\}.$$

In the case $\partial V = \{a, b\}$, $\Delta h = 0$ on V° , $h(a) = 1$ and $h(b) = 0$, then

$$(44) \quad \mathcal{E}(h) = \min\{\mathcal{E}(f) : f(a) = 1, f(b) = 0\} = \mathbb{C}_{ab}.$$

The following proof that the Dirichlet minimiser is harmonic is analogous to the P.D.E. situation.²⁰ The proof that the Dirichlet energy of the harmonic function is $\mathbb{C}_{ab}$ follows essentially directly from Green's Identity and the definition of $\mathbb{C}_{ab}$.

Proof. Suppose $\phi = 0$ on ∂V . Then

$$\begin{aligned}
\mathcal{E}(h + \phi, h + \phi) &= \mathcal{E}(h, h) + 2\mathcal{E}(h, \phi) + \mathcal{E}(\phi, \phi) \quad (\text{bilinearity}) \\
&= \mathcal{E}(h, h) - 2(\Delta h, \phi) + \mathcal{E}(\phi, \phi) \quad (\text{Green's identity}) \\
&= \mathcal{E}(h, h) + \mathcal{E}(\phi, \phi) \quad (\text{since } \Delta h = 0 \text{ on } V^\circ \text{ and } \phi = 0 \text{ on } \partial V) \\
&\geq \mathcal{E}(h, h)
\end{aligned}$$

Equality holds in the last line iff ϕ is constant, i.e. $\phi \equiv 0$.

This shows that h is the unique minimiser.

In the second case, by Green's identity (41), $\mathcal{E}(h) = -c_a \Delta h(a)$. But $-c_a \Delta h(a)$ is the current i_a under the voltage h , by (6). Since $h(a) = 1$ and $h(b) = 0$, it follows $i_a = \mathbb{C}_{ab}$ by the definition of $\mathbb{C}_{ab}$ in (10). □

7.4. Flows. The following definitions are motivated by the properties of a current induced by a voltage applied on ∂V , over a network with possibly unknown conductances.

Fix $\partial V \subset V$. A *flow* is a function j on the directed edges of $\mathbb{G}$ such that

$$\begin{aligned}
(45) \quad & j_{xy} = 0 \text{ for } x \not\sim y \\
& j_{xy} = -j_{yx} \\
& \sum_y j_{xy} = 0 \text{ if } x \in V^\circ.
\end{aligned}$$

²⁰(See footnote 16.) The harmonic extension to Ω of $h : \partial\Omega \rightarrow \mathbb{R}$ is the unique minimiser in $H^1(\Omega)$ of $\mathcal{E}(f) := \frac{1}{2} \int_\Omega |Df|^2$ with the prescribed boundary conditions. Existence comes from taking a minimising sequence and standard P.D.E. arguments to show the sequence converges in the H^1 sense.

For here, just note that if h is harmonic and $\phi \in H_0^1(\Omega)$ then

$$\begin{aligned}
\mathcal{E}(h + \phi) &= \frac{1}{2} \int_\Omega |Dh|^2 + \int_\Omega Dh \cdot D\phi + \frac{1}{2} \int_\Omega |D\phi|^2 \\
&= \mathcal{E}(h) - \int_\Omega \Delta h \cdot \phi + \mathcal{E}(\phi) \\
&= \mathcal{E}(h) + \mathcal{E}(\phi) \\
&\geq \mathcal{E}(h), \text{ with equality iff } \phi = 0 \text{ a.e.}
\end{aligned}$$

So h is indeed the unique minimiser.

The *flow out of* x for $x \in V$ is

$$j_x := \sum_y j_{xy}.$$

Hence $j_x = 0$ if $x \in V^\circ$.

The *energy* of j is

$$(46) \quad \mathcal{E}(j) := \frac{1}{2} \sum_{x,y} j_{xy}^2 c_{xy}^{-1}.$$

This is the same as the energy of v if j is the current corresponding to the voltage v .

If $\partial V = \{a, b\}$ and $j_a = -j_b = 1$, then j is a *unit flow from a to b* .

7.5. Properties of Flows. Suppose j is a flow and $w : V \rightarrow \mathbb{R}$. Then for any function w_x ,

$$(47) \quad \sum_{x \in \partial V} j_x = 0, \quad j_a = -j_b \text{ if } \partial V = \{a, b\},$$

$$\frac{1}{2} \sum_{x,y} (w_x - w_y) j_{xy} = \sum_{x \in \partial V} w_x j_x.$$

Note the latter is just Green's identity (41) if $w = f$ and j is the current corresponding to the voltage g .

Proof. The first line follows from the second with $w_x \equiv 1$.

For the second,

$$\begin{aligned} \frac{1}{2} \sum_{x,y} (w_x - w_y) j_{xy} &= \frac{1}{2} \sum_{x,y} w_x j_{xy} - \frac{1}{2} \sum_{x,y} w_y j_{xy} = \frac{1}{2} \sum_{x,y} w_x j_{xy} - \frac{1}{2} \sum_{x,y} w_x j_{yx} \\ &= \sum_{x,y} w_x j_{xy} = \sum_x w_x j_x \\ &= \sum_{x \in \partial V} w_x j_x. \end{aligned}$$

□

7.6. Thompson's Second Principle.

Proposition 7.2. Suppose i is the unit flow corresponding to a voltage v applied to a and b , with $v(b) = 0$. Then

$$(48) \quad \mathcal{E}(i) = \min\{\mathcal{E}(j) : j \text{ is a unit flow from } a \text{ to } b\} = \mathbb{R}_{ab}.$$

The proof of the following is analogous to that for Thompson's first principle. The value of the minimiser follows from that in the first principle by a simple scaling argument.

Proof. For feasible j let $j_{xy} = i_{xy} + d_{xy}$. Then d_{xy} is a flow with $d_a = d_b = 0$. Hence

$$\begin{aligned} \mathcal{E}(j) &= \frac{1}{2} \sum_{x,y} (i_{xy} + d_{xy})^2 r_{xy} \\ &= \frac{1}{2} \sum_{x,y} i_{xy}^2 r_{xy} + \sum_{x,y} i_{xy} d_{xy} r_{xy} + \frac{1}{2} \sum_{x,y} d_{xy}^2 r_{xy}. \end{aligned}$$

From (47) the second term on the right equals

$$\sum_{x,y} (v(x) - v(y)) d_{xy} = 2v(a) d_a + 2v(b) d_b = 0.$$

Hence

$$\mathcal{E}(j) \geq \mathcal{E}(i),$$

with equality iff $d = 0$, i.e. $i = j$.

If a unit current flows from a to b the applied voltage is $\mathbb{R}_{ab}$, by the definition of $\mathbb{R}_{ab}$. But if the applied voltage is 1 then the Dirichlet energy is $\mathbb{C}_{ab}$ from Thompson's first principle. Hence

if the applied voltage is $\mathbb{R}_{ab}$ the Dirichlet energy is multiplied by $\mathbb{R}_{ab}^2$ from the definition (39) of Dirichlet energy, since all voltages will then also be scaled up by $\mathbb{R}_{ab}$. That is,

$$\mathcal{E}(i) = \mathbb{R}_{ab}^2 \mathbb{C}_{ab} = \mathbb{R}_{ab}.$$

□

7.7. Alternative expression for Dirichlet energy. It is convenient for calculations to write $\mathcal{E}(f, g)$ as a bilinear form in the standard manner, using a positive semi definite matrix. This will give a slightly different, but more useful, way of expressing the formula for harmonic extensions and for Green's function. In particular, we do this in Section 10 concerning trace networks.

Note:

- (1) $\mathcal{E}(f, g) = \frac{1}{2} \sum_{x,y} c_{xy} (f_x - f_y)(g_x - g_y)$ defines a positive semi definite symmetric bilinear form, and $\mathcal{E}(f, f) = 0$ iff f constant.²¹
- (2) The terms c_{xx} are zero. Replacing c_{xx} by any other value will not alter the value of $\mathcal{E}(f, g)$.
- (3) As for any (positive semi definite) symmetric bilinear form, $\mathcal{E}(f, g) = f^T B g = g^T B f$ for some (positive semi definite) symmetric matrix B .
- (4) Formula (50) expresses the energy as a sum of contributions from *nodes* rather than edges.

In the following note that $-A$ is the positive semi definite matrix.

Proposition 7.3. Let $A = (a_{xy})$ for $x, y \in V$, where

$$(49) \quad a_{xy} = \begin{cases} c_{xy} & x \neq y \\ -c_x & x = y \end{cases}.$$

Then

$$(50) \quad \boxed{\mathcal{E}(f, g) = - \sum_{x,y} a_{xy} f_x g_y} \quad (= -f^T A g = -g^T A f).$$

Proof.

$$\begin{aligned} \mathcal{E}(f, g) &= \sum_{x,y} c_{xy} f_x g_x - \sum_{x,y} c_{xy} f_x g_y \quad (\text{since } c_{xy} = c_{yx}) \\ &= \sum_x c_x f_x g_x - \sum_{\substack{x,y \\ x \neq y}} c_{xy} f_x g_y \quad (\text{since } c_x = \sum_y c_{xy} \text{ and } c_{xx} = 0) \\ &= - \sum_x a_{xx} f_x g_x - \sum_{\substack{x,y \\ x \neq y}} a_{xy} f_x g_y \\ &= - \sum_{x,y} a_{xy} f_x g_y. \end{aligned}$$

□

7.8. Cuts and Shorts.

²¹Think of the graph of $\mathcal{E}(f, f)$ as an $n - 1$ dimensional parabola $\times$ straight line, sitting on $\mathbb{R}^n$, where n is the cardinality of V .

8. Matrices

8.1. Notation. Consider $G \subset V$ and let $H = V \setminus G$.

Typically one has a sequence of vertices $(V_n)_{n \geq 0}$ of prefractal approximations to a fractal. One takes $V = V_n$ and either $G = V_{n-1}$, or V_0 , or a for some $a \in V_0$, or some subset of V_0 .

For $f : V \rightarrow \mathbb{R}$ let f_G and f_H denote the restrictions to G and H respectively.

Write

$$(51) \quad A = \begin{bmatrix} A_{GG} & A_{GH} \\ A_{HG} & A_{HH} \end{bmatrix},$$

with an obvious notation.

Let D be the *diagonal matrix* $D_{xy} = c_x d_{xy}$. Let D_G and D_H be the restrictions of D to G and H respectively.

Let P be the standard *transition matrix* $P_{xy} = c_{xy}/c_x$.

Then the *Laplacian* is

$$(52) \quad \Delta = P - I.$$

It follows²²

$$(53) \quad A = D(P - I) = D \Delta.$$

Note that A is symmetric, but Δ and P are not.

8.2. Harmonic Extensions.

Proposition 8.1. *Suppose $f_G = g$ is prescribed and $f_H = h$ is the extension to H such that f is harmonic on H . Then*

$$(54) \quad h = -A_{HH}^{-1} A_{HG} g.$$

In this case,

$$(55) \quad -\mathcal{E}(f) = g^T (A_{GG} - A_{GH} A_{HH}^{-1} A_{HG}) g.$$

Proof. Proceed in the standard manner.²³ (Alternatively, since we know that the minimiser is harmonic we can also argue directly as in the Remark following the proof.)

$$(56) \quad \begin{aligned} -\mathcal{E}(f) &= \begin{bmatrix} g^T & h^T \end{bmatrix} \begin{bmatrix} A_{GG} & A_{GH} \\ A_{HG} & A_{HH} \end{bmatrix} \begin{bmatrix} g^T \\ h^T \end{bmatrix} \\ &= g^T A_{GG} g + g^T A_{GH} h + h^T A_{HG} g + h^T A_{HH} h \quad (\text{sum of scalars}) \\ &= g^T A_{GG} g + 2g^T A_{GH} h + h^T A_{HH} h \quad (\text{take transpose of previous third term}). \end{aligned}$$

Since the minimum of $\mathcal{E}(f)$ occurs at h , replace h by $h + t\phi$, differentiate w.r.t. t , and set $t = 0$ to get, $\forall \phi : H \rightarrow \mathbb{R}$,

$$0 = 2g^T A_{GH} \phi + h^T A_{HH} \phi + \phi^T A_{HH} h = 2g^T A_{GH} \phi + 2h^T A_{HH} \phi.$$

It follows $g^T A_{GH} + h^T A_{HH} = 0$, so $A_{HG} g + A_{HH} h = 0$, and so

$$h = -A_{HH}^{-1} A_{HG} g.$$

Substituting into (56) gives

$$\begin{aligned} -\mathcal{E}(f) &= g^T (A_{GG} - 2g^T A_{GH} A_{HH}^{-1} A_{HG} + A_{GH} A_{HH}^{-1} A_{HH} A_{HH}^{-1} A_{HG}) g \\ &= g^T (A_{GG} - A_{GH} A_{HH}^{-1} A_{HG}) g. \end{aligned}$$

□

²²Note that for any matrix B with the same number of rows as D , DB is obtained from B by multiplying all terms in row i by D_{ii} . Similarly, if B has the same number of columns as D then BD is obtained from B by multiplying all terms in column j by D_{jj} .

²³Consider the minimum of $ag^2 + 2bgh + ch^2$ for a fixed scalar g and variable scalar h . The derivative is zero at $2bg + 2ch = 0$. Substituting $h = -c^{-1}b$ the minimum value is $(a - b^2c^{-1})g^2$.

Remark: We can also obtain (54) by using the fact h is harmonic. That is, from (53) the equations for the harmonic extension $f = \begin{bmatrix} h \\ g \end{bmatrix}$ of g can be written

$$\begin{bmatrix} I_G & O_{GH} \\ D_H^{-1}A_{HG} & D_H^{-1}A_{HH} \end{bmatrix} \begin{bmatrix} g \\ h \end{bmatrix} = \begin{bmatrix} g \\ 0 \end{bmatrix}$$

Hence

$$\begin{aligned} D_H^{-1}(A_{HG}g + A_{HH}h) &= 0 \\ \therefore h &= -A_{HH}^{-1}A_{HG}g. \end{aligned}$$

8.3. Green's Function.

Proposition 8.2. *Fix $a \in H$. As in (25) the Green's function corresponding to the boundary G and centred at a is defined to be the unique solution of*

$$(57) \quad \begin{aligned} -\Delta \mathcal{G}_a(x) &= \frac{1}{c_a} \chi_a(x) & x \in G \\ \mathcal{G}_a(x) &= 0 & x \in H. \end{aligned}$$

Let $\mathcal{G}$ be the square matrix whose a th column is given by $\mathcal{G}_a(x)$ for $x \in H$. Then

$$(58) \quad \mathcal{G} = -A_{HH}^{-1}.$$

Proof. Restricting $\mathcal{G}_a$ to a vector on H ,

$$\begin{aligned} (I_H - P_{HH})\mathcal{G}_a &= \frac{1}{c_a} \chi_a \quad (\text{this uses } \mathcal{G}_a(x) = 0 \text{ if } x \in G) \\ (I_H - P_{HH})\mathcal{G} &= D_H^{-1} \quad (\text{since } a \in H \text{ is arbitrary}) \\ \mathcal{G} &= (I_H - P_{HH})^{-1}D_H^{-1} = -A_{HH}^{-1}. \end{aligned}$$

□

8.4. Other Quantities.

- (1) *Resistance Metric:* Let $G = \{y\}$. Then from Section ??, $\mathbb{R}_{xy} = \mathcal{G}_x^{\{y\}}(x)$, and so from (58),

$$(59) \quad \boxed{\mathbb{R}_{xy} = -(A_{HH}^{-1})_{xx} \text{ if } G = \{y\}.}$$

- (2) *Visit Times:* Let $\tilde{P}$ be the transition matrix for the RW with *absorbing states* G . Then clearly $-D^{-1}A_{HH} = I_{HH} - \tilde{P}$ and so

$$-A_{HH}^{-1}D = (I_{HH} - \tilde{P})^{-1} = I_{HH} + \tilde{P} + \tilde{P}^2 + \dots + \tilde{P}^k + \dots$$

Since $\tilde{P}_{xy}^k = P^x(\text{RW is at } y \text{ after } k \text{ steps})$, the xy element of the right side is $\mathbb{E}^x V_{y \rightarrow G}$. Hence

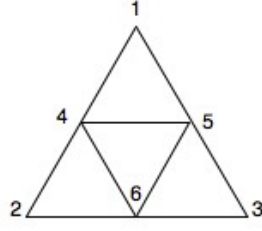
$$(60) \quad \boxed{\mathbb{E}^x V_{y \rightarrow G} = -c_y (A_{HH}^{-1})_{xy}.}$$

- (3) *Crossing Times:* Since $\mathbb{E}^x T_G = \sum_y \mathbb{E}^x V_{y \rightarrow G}$,

$$(61) \quad \boxed{\mathbb{E}^x T_G = -\sum_y c_y (A_{HH}^{-1})_{xy}.}$$

9. Matrix Examples

All computations done in Maple. See accompanying worksheet.


 FIGURE 4. The triangle T and the $SG(2)$ prefractal

9.1. Simple Triangle T. Consider the triangle network $\{1, 2, 3\}$ with unit resistances in Figure 4.

Then

$$A = \begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix}$$

With $G = \{1\}$,

$$-A_{HH}^{-1} = \begin{bmatrix} 2/3 & 1/3 \\ 1/3 & 2/3 \end{bmatrix}$$

Going down the diagonal, the effective resistances are

$$\mathbb{R}_{12} = \mathbb{R}_{13} = 2/3.$$

The expected crossing times from x to 1 are obtained by multiplying the elements of column x by the degrees of the elements and summing:

$$\mathbb{E}^2 T_1 = \mathbb{E}^3 T_1 = 2.$$

With $G = \{1, 2\}$,

$$-A_{HH}^{-1} = \frac{1}{2}.$$

So

$$\mathbb{E}_{\{1,2\}3} = \frac{1}{2}, \quad \mathbb{E}^3 T_{\{1,2\}} = 2 \times \frac{1}{2} = 1,$$

as is clear anyway.

The crossing time theorem $\mathbb{E}^1 T_2 + \mathbb{E}^2 T_1 = \mathbb{R}_{12} C(\mathbb{G}) = 4$ is satisfied.

9.2. SG(2) prefractal. The network is shown in Figure 4. Each of the 6 edges is given unit resistance.

Then

$$A = \begin{bmatrix} -2 & 0 & 0 & 1 & 1 & 0 \\ 0 & -2 & 0 & 1 & 0 & 1 \\ 0 & 0 & -2 & 0 & 1 & 1 \\ 1 & 1 & 0 & -4 & 1 & 1 \\ 1 & 0 & 1 & 1 & -4 & 1 \\ 0 & 1 & 1 & 1 & 1 & -4 \end{bmatrix},$$

We consider the three cases $G = \{1\}, \{1, 2\}, \{1, 2, 3\}$. Then one computes respectively:

$$-A_{HH}^{-1} = \begin{bmatrix} \frac{10}{9} & 5/9 & 5/9 & 4/9 & 2/3 \\ 5/9 & \frac{10}{9} & 4/9 & 5/9 & 2/3 \\ 5/9 & 4/9 & \frac{11}{18} & \frac{7}{18} & 1/2 \\ 4/9 & 5/9 & \frac{7}{18} & \frac{11}{18} & 1/2 \\ 2/3 & 2/3 & 1/2 & 1/2 & 5/6 \end{bmatrix}$$

$$\begin{bmatrix} 5/6 & 1/6 & 1/3 & 1/3 \\ 1/6 & 1/3 & 1/6 & 1/6 \\ 1/3 & 1/6 & \frac{13}{30} & \frac{7}{30} \\ 1/3 & 1/6 & \frac{7}{30} & \frac{13}{30} \end{bmatrix} \quad \begin{bmatrix} 3/10 & 1/10 & 1/10 \\ 1/10 & 3/10 & 1/10 \\ 1/10 & 1/10 & 3/10 \end{bmatrix}$$

In particular

$$\mathbb{R}_{12} = \mathbb{R}_{13} = \frac{10}{9} \text{ (first matrix, first 2 diagonal elements)}$$

$$\mathbb{E}^2 T_1 = \mathbb{E}^3 T_1 = 10 \text{ (first matrix, first two columns } \times \text{ corresponding conductances)}$$

$$\mathbb{R}_{\{1,2\}}^3 = \frac{5}{6} \text{ (second matrix, first diagonal element)}$$

$$\mathbb{E}^3 T_{\{1,2\}} = 5 \text{ (second matrix, first column } \times \text{ corresponding conductances)}$$

$$C(\mathbb{G}) = 3 \times 2 + 3 \times 4 = 18.$$

The crossing time theorem $\mathbb{E}^2 T_1 + \mathbb{E}^1 T_2 = \mathbb{R}_{12} C(\mathbb{G}) = 20$ is again satisfied.

Mass, Time and Resistance Scaling: Since edges all have unit resistance, i.e. the network is simple, the mass is $\sum_x \text{degree}(x)$, or equivalently twice the number of edges, or equivalently 6 times the number of triangles, i.e. 18. But the mass of T is 6. So the mass scaling m is $18/6 = 3$.

Next compare the crossing times and effective resistances above with those for T on page ?? . Let τ and ρ , respectively, denote the factors by which the walk times and resistance between comparable nodes scale up in passing from T to $SG(2)$.

Then $m = 18/6 = 3$, $\tau = 10/2$ or $5/1 = 5$, and $\rho = \frac{10}{9} / \frac{2}{3}$ or $\frac{5}{6} / \frac{1}{2} = \frac{5}{3}$. That is:

$$(62) \quad \boxed{m = 3, \tau = 5, \rho = 5/3.}$$

Note the Einstein relation $\tau = m \times \rho$ holds.

9.3. SG(3) prefractal. The network is shown in Figure 5. Each of the 18 edges is given unit resistance.

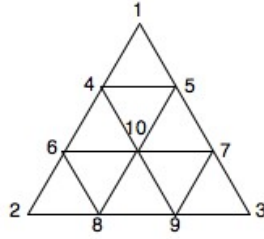


FIGURE 5. The $SG(3)$ prefractal

We consider the cases $G = \{1\}, \{1, 2\}$. Then one computes:

$$A = \begin{bmatrix} -2 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & -4 & 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & -4 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & -4 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & -4 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & -4 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & -4 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & -6 \end{bmatrix},$$

and

$$A_{HH}^{-1} = \begin{bmatrix} \frac{10}{7} & 5/7 & \frac{11}{21} & \frac{10}{21} & \frac{19}{21} & 2/3 & \frac{20}{21} & \frac{16}{21} & 5/7 \\ 5/7 & \frac{10}{7} & \frac{10}{21} & \frac{11}{21} & 2/3 & \frac{19}{21} & \frac{16}{21} & \frac{20}{21} & 5/7 \\ \frac{11}{21} & \frac{10}{21} & \frac{17}{28} & \frac{11}{28} & \frac{15}{28} & \frac{13}{28} & \frac{43}{84} & \frac{41}{84} & 1/2 \\ \frac{10}{21} & \frac{11}{21} & \frac{11}{28} & \frac{17}{28} & \frac{13}{28} & \frac{15}{28} & \frac{41}{84} & \frac{43}{84} & 1/2 \\ \frac{19}{21} & 2/3 & \frac{15}{28} & \frac{13}{28} & \frac{83}{84} & \frac{53}{84} & \frac{23}{28} & \frac{59}{84} & \frac{29}{42} \\ 2/3 & \frac{19}{21} & \frac{13}{28} & \frac{15}{28} & \frac{53}{84} & \frac{83}{84} & \frac{59}{84} & \frac{23}{28} & \frac{29}{42} \\ \frac{20}{21} & \frac{16}{21} & \frac{43}{84} & \frac{41}{84} & \frac{23}{28} & \frac{59}{84} & \frac{13}{12} & \frac{23}{28} & \frac{31}{42} \\ \frac{16}{21} & \frac{20}{21} & \frac{41}{84} & \frac{43}{84} & \frac{59}{84} & \frac{23}{28} & \frac{23}{28} & \frac{13}{12} & \frac{31}{42} \\ 5/7 & 5/7 & 1/2 & 1/2 & \frac{29}{42} & \frac{29}{42} & \frac{31}{42} & \frac{31}{42} & \frac{17}{21} \end{bmatrix},$$

$$\begin{bmatrix} \frac{15}{14} & 3/14 & 2/7 & 3/14 & 4/7 & 2/7 & 4/7 & \frac{5}{14} \\ 3/14 & \frac{523}{1260} & \frac{55}{252} & \frac{257}{1260} & \frac{277}{1260} & \frac{41}{252} & \frac{263}{1260} & \frac{5}{21} \\ 2/7 & \frac{55}{252} & \frac{113}{252} & \frac{41}{252} & \frac{79}{252} & \frac{43}{252} & \frac{65}{252} & \frac{11}{42} \\ 3/14 & \frac{257}{1260} & \frac{41}{252} & \frac{523}{1260} & \frac{263}{1260} & \frac{55}{252} & \frac{277}{1260} & \frac{5}{21} \\ 4/7 & \frac{277}{1260} & \frac{79}{252} & \frac{263}{1260} & \frac{853}{1260} & \frac{65}{252} & \frac{587}{1260} & \frac{5}{14} \\ 2/7 & \frac{41}{252} & \frac{43}{252} & \frac{55}{252} & \frac{65}{252} & \frac{113}{252} & \frac{79}{252} & \frac{11}{42} \\ 4/7 & \frac{263}{1260} & \frac{65}{252} & \frac{277}{1260} & \frac{587}{1260} & \frac{79}{252} & \frac{853}{1260} & \frac{5}{14} \\ \frac{5}{14} & \frac{5}{21} & \frac{11}{42} & \frac{5}{21} & \frac{5}{14} & \frac{11}{42} & \frac{5}{14} & \frac{19}{42} \end{bmatrix},$$

respectively.

Multiplying by the conductance vectors $[2 \ 2 \ 4 \ 4 \ 4 \ 4 \ 4 \ 4 \ 6]$ and $[2 \ 4 \ 4 \ 44 \ 4 \ 4 \ 6]$ respectively, we get the expected transition time vectors

$$\left[\frac{180}{7} \ \frac{180}{7} \ 17 \ 17 \ \frac{167}{7} \ \frac{167}{7} \ \frac{179}{7} \ \frac{179}{7} \ \frac{162}{7} \right] \text{ and } \left[\frac{90}{7} \ \frac{53}{7} \ \frac{59}{7} \ \frac{53}{7} \ \frac{83}{7} \ \frac{59}{7} \ \frac{83}{7} \ \frac{72}{7} \right].$$

In particular

$$\begin{aligned} \mathbb{R}_{12} &= \mathbb{R}_{13} = \frac{10}{7} \text{ (first } A_{HH}^{-1}, \text{ first 2 diagonal elements)} \\ \mathbb{E}^2 T_1 &= \mathbb{E}^3 T_1 = 180/7 \text{ (first 2 entries of first transition vector)} \\ \mathbb{R}_{\{1,2\}}^3 &= \frac{15}{14} \text{ (second } A_{HH}^{-1}, \text{ first diagonal element)} \\ \mathbb{E}^3 T_{\{1,2\}} &= \frac{90}{7} \text{ (first entry of second transition vector)} \\ C(\mathbb{G}) &= 3 \times 2 + 6 \times 4 + 6 = 36. \end{aligned}$$

The crossing time theorem $\mathbb{E}^2 T_1 + \mathbb{E}^1 T_2 = \mathbb{R}_{12} C(\mathbb{G}) = 360/7$ is again satisfied.

Mass, Time and Resistance Scaling: (Compare with $SG(2)$ for notation.) For $SG(3)$ we obtain the following mass, time and resistance scalings: $m = 36/6 = 6$, $\tau = \frac{180}{7}/2$ or $\frac{90}{7}/1 = \frac{90}{7}$ and $\rho = \frac{10}{7}/\frac{2}{3}$ or $\frac{15}{14}/\frac{1}{2} = \frac{15}{7}$. That is:

$$(63) \quad \boxed{m = 6, \tau = 90/7, \rho = 15/7.}$$

Note the Einstein relation $\tau = m \times \rho$ holds.

10. Trace Networks

10.1. Motivation. Consider the usual sequence of prefractal approximations to $SG(2)$ or $SG(3)$. Write this as

$$G_0 \subset G_1 \subset G_2 \subset \cdots \subset G_* := \bigcup_{n \geq 0} G_n.$$

Beginning with conductances 1 on the three edges of G_0 we want conductances on the edges of each G_n which give a sequence of compatible networks. That is, we require each G_n to be compatible with G_{n-1} , which means that if G_n is restricted to V_{n-1} it should give G_{n-1} in the black box sense of electrical networks, see Section 10.2. In fact, the conductance on each edge of G_n must be ρ^n , where $\rho = 5/3$ or $15/7$ for $SG(2)$ or $SG(3)$ respectively, and is the resistance scaling factor on page 20 and 22.

To see this, suppose initially that both G_0 and G_1 have unit resistance on each edge. Then from the discussion at the end of Sections 9.2 and 9.3 the effective resistances on V_0 induced from G_1 are $\rho \times$ the corresponding effective resistance on G_0 . So to keep the same effective resistances on V_0 we need the resistance on each edge of G_1 to be ρ^{-1} and hence the conductance to be ρ . In a similar manner, the conductance on each edge of G_n should be ρ^n .

Note if conductances are multiplied by ρ then resistances are multiplied by ρ^{-1} and masses (obtained from the measure (c_x)) are multiplied by ρ . In particular the Einstein relation “time” = “mass” $\times$ “resistance” is preserved.

10.2. The Black Box Problem. Suppose we can apply arbitrary voltages at the various nodes in V and can then measure the current in or out of any node. This treats the network as a “black box” in which we cannot directly access the wiring. In particular, we cannot measure the current across individual edges.

However, this “input-output” information is sufficient to reconstruct the network, that is find the conductances c_{xy} .

Note that this is different from finding the *effective* conductances $\mathbb{C}_{xy}$. In this case one applies voltage 1 at x and 0 at y . The current flow into the network through x or equivalently out of the network through y , is $\mathbb{C}_{xy}$.

To find c_{xy} apply voltage 1 at x and 0 at *all* other z , including y . Then the current flow along xy is clearly c_{xy} . This is also the (*measurable*) flow out of the network through y since there is no current flow between y and any node other than x . Once we know the c_{xy} we can find i_{xy} by measuring voltage differences and using Ohm’s law.

Note also that the energy functional completely determines the conductances (and is trivially determined by the conductances), since for $x \neq y$, $c_{xy} = -\mathcal{E}(\delta_x, \delta_y)$ from (50).

Finally, the energy form determines the effective conductances, and the converse is true by induction on the number of vertices. (This is a bit of a calculation, see [Ki, p47]. Is there a better way?)

To summarise, one has the following information content equivalences:

$$(64) \quad \text{input-output info.} \iff \text{conductances} \iff \text{Dirichlet form} \iff \text{effective conductances.}$$

10.3. Induced Networks.